

Please check the examination details below before entering your candidate information

Candidate surname					Other names				
Centre Number					Candidate Number				

Pearson Edexcel International GCSE

Time 2 hours	Paper reference	4PM1/01
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Further Pure Mathematics
PAPER 1



Calculators may be used.	Total Marks
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Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Without sufficient working, correct answers may be awarded no marks.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You must **NOT** write anything on the formulae page.
Anything you write on the formulae page will gain NO credit.

Information

- The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Check your answers if you have time at the end.

Turn over ►

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International GCSE in Further Pure Mathematics Formulae sheet

Mensuration

Surface area of sphere = $4\pi r^2$

Curved surface area of cone = $\pi r \times$ slant height

Volume of sphere = $\frac{4}{3}\pi r^3$

Series

Arithmetic series

Sum to n terms, $S_n = \frac{n}{2}[2a + (n-1)d]$

Geometric series

Sum to n terms, $S_n = \frac{a(1-r^n)}{(1-r)}$

Sum to infinity, $S_\infty = \frac{a}{1-r} \quad |r| < 1$

Binomial series

$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots + \frac{n(n-1)\dots(n-r+1)}{r!}x^r + \dots \quad \text{for } |x| < 1, n \in \mathbb{Q}$

Calculus

Quotient rule (differentiation)

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$$

Trigonometry

Cosine rule

In triangle ABC : $a^2 = b^2 + c^2 - 2bc \cos A$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$\sin(A-B) = \sin A \cos B - \cos A \sin B$$

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

Logarithms

$$\log_a x = \frac{\log_b x}{\log_b a}$$

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Answer all TEN questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

- 1 Given that $\frac{2\sqrt{3}-4}{3\sqrt{3}+5}$ can be written in the form $a + b\sqrt{3}$ where a and b are integers, find, without using a calculator, the value of a and the value of b
- Show your working clearly.

(3)

$$\frac{2\sqrt{3}-4}{3\sqrt{3}+5} \cdot \frac{3\sqrt{3}-5}{3\sqrt{3}-5} = \frac{18-22\sqrt{3}+20}{27-25}$$

$$= \frac{38-22\sqrt{3}}{2} = 19-11\sqrt{3}$$

(Total for Question 1 is 3 marks)



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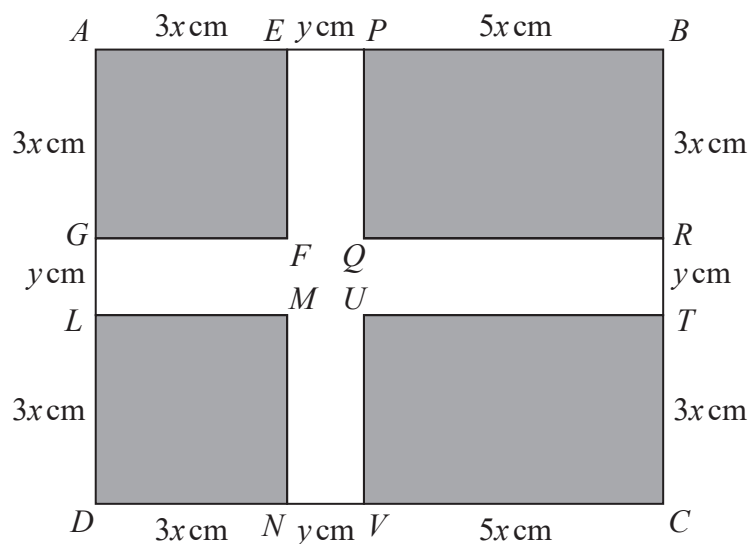


Figure 1

Figure 1 shows the design for a flag consisting of a white cross on a grey background.

$AEFG$ and $DLMN$ are squares with sides of length $3x$ cm.

$BPQR$ and $CTUV$ are rectangles with sides of length $5x$ cm and $3x$ cm.

The width of the cross is y cm.

The total area of the flag is H cm²

(a) Write down an expression, in terms of x and y , for H .

(1)

Given that the area of the cross is K cm²

(b) show that $K = 14xy + y^2$

(3)

The total area of the flag is to be 3432 cm² and the area of the cross is to be 1080 cm²

(c) Find the value of x and the value of y

(5)

$$(a) H = (8x + y)(6x + y) = 48x^2 + 14xy + y^2$$

$$(b) K = H - 2 \cdot 9x^2 - 2 \cdot 15x^2$$

$$K = H - 48x^2$$

$$K = 14xy + y^2$$



Question 2 continued

$$(c) \quad H = 48x^2 + 14xy + y^2 = 3432$$
$$K = 14xy + y^2 = 1080$$

$$\Rightarrow 48x^2 = 2352$$

$$x^2 = 49$$

$$x = 7$$

$$\sim) K = 98y + y^2 - 1080 = 0$$

$$y^2 + 98y - 1080 = 0$$

$$(y + 108)(y - 10) = 0$$

$$y = -108$$

$$y = 10$$

rej



$$\sin \theta = \frac{\text{opp}}{\text{hyp}}$$

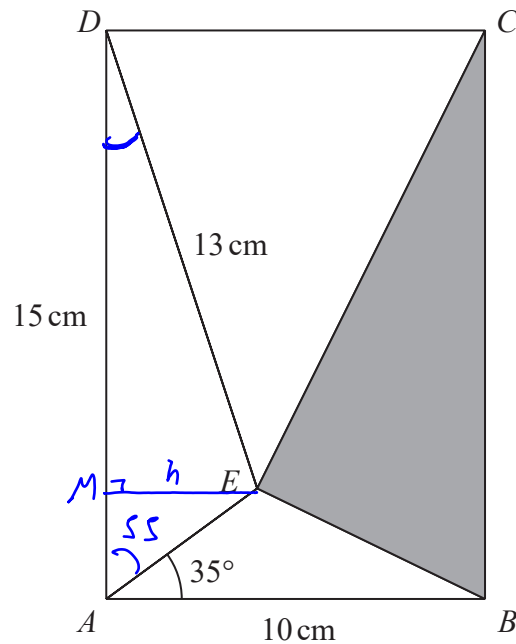


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Figure 2

Figure 2 shows a rectangle $ABCD$ with $AB = 10$ cm and $AD = 15$ cm.
 E is the point inside the rectangle such that $DE = 13$ cm and angle $BAE = 35^\circ$

Given that angle AED is obtuse,

find the area, in cm^2 to one decimal place, of triangle BCE .

(7)

$$\hat{DAE} = 90 - 35 = 55$$

Sine rule:

$$\frac{\sin E}{15} = \frac{\sin 55}{13} \Rightarrow \sin E = \frac{15 \sin 55}{13} \Rightarrow E = 70.94$$

$$\hat{AED} = 180 - 70.94 = 109.$$

$$\hat{ADE} = 180 - 55 - 109 = 16$$

$$\sin 16 = \frac{h}{13} \Rightarrow h = 13 \sin 16 = 3.58$$

$$A = \frac{15 \cdot (10 - 3.58)}{2} = 48.2$$



- 4 The common ratio of a geometric series G is positive.

The sum of the first 4 terms of G is 80

The sum to infinity of G is 81

Show that the sum of the first 7 terms of G differs from the sum to infinity of G by $\frac{1}{27}$

(7)

$$S_4 = \frac{a(1-r^4)}{1-r} = 80$$

$$S_{\infty} = \frac{a}{1-r} = 81 \Rightarrow a = 81(1-r)$$

$$\Rightarrow \frac{81(1-r)(1-r^4)}{(1-r)} = 80$$

$$\Rightarrow (1-r^4) = \frac{80}{81} \Rightarrow -r^4 = \frac{80}{81} - 1$$

$$\Rightarrow r^4 = \frac{1}{81}$$

$$r = \pm \frac{1}{3} \sim r = +\frac{1}{3}$$

$$\sim a = 54$$

$$S_7 = \frac{54(1-(\frac{1}{3})^7)}{1-\frac{1}{3}} = \frac{2186}{27}$$

$$S_{\infty} - S_7 = 81 - \frac{2186}{27} = \frac{1}{27}$$



- 5 Given that $(2 + 3x)^{-1}$ can be expressed in the form $p(1 + qx)^{-1}$ where p and q are constants,

(a) find the value of p and the value of q

(2)

(b) Hence expand $(2 + 3x)^{-1}$ in ascending powers of x up to and including the term in x^3 , expressing each coefficient as an exact fraction in its lowest terms.

(3)

$$f(x) = \frac{1+x}{2+3x}$$

(c) Obtain a series expansion for $f(x)$, in ascending powers of x up to and including the term in x^3 , expressing each coefficient as an exact fraction in its lowest terms.

(2)

(d) Hence use algebraic integration to obtain an estimate, to 4 decimal places, of

$$\int_0^{0.5} f(x) \, dx$$

(4)

$$(a) \quad 2^{-1} \left(1 + \frac{3}{2}x\right)^{-1} = \frac{1}{2} \left(1 + \frac{3}{2}x\right)^{-1}$$

$$\Rightarrow p = \frac{1}{2} \quad q = \frac{3}{2}$$

$$(b) \quad (2+3x)^{-1} = \frac{1}{2} \left(1 + \frac{3}{2}x\right)^{-1}$$

$$= \frac{1}{2} \left[1 - \frac{3}{2}x + \frac{(-1)(-2)}{2!} \left(\frac{3}{2}x\right)^2 + \frac{(-1)(-2)(-3)}{3!} \left(\frac{3}{2}x\right)^3 \right]$$

$$= \frac{1}{2} \left[1 - \frac{3}{2}x + \frac{9}{4}x^2 - \frac{27}{8}x^3 \right]$$

$$= \frac{1}{2} - \frac{3}{4}x + \frac{9}{8}x^2 - \frac{27}{16}x^3$$

$$(c) \quad f(x) = (1+x)(2+3x)^{-1} = (1+x) \left(\frac{1}{2} - \frac{3}{4}x + \frac{9}{8}x^2 - \frac{27}{16}x^3 \right)$$

$$= \frac{1}{2} - \frac{3}{4}x + \frac{9}{8}x^2 - \frac{27}{16}x^3 + \frac{1}{2}x - \frac{3}{4}x^2 + \frac{9}{8}x^3 + \dots$$

$$= \frac{1}{2} - \frac{1}{4}x + \frac{3}{8}x^2 - \frac{9}{16}x^3$$



Question 5 continued

$$(2) \int_0^{\frac{1}{2}} f(x) dx = \int_0^{\frac{1}{2}} \frac{1}{2} - \frac{1}{4}x + \frac{3}{8}x^2 - \frac{9}{16}x^3 dx$$

$$= \left[\frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{8}x^3 - \frac{9}{64}x^4 \right]_0^{\frac{1}{2}}$$

$$= \frac{1}{4} - \frac{1}{32} + \frac{1}{64} - \frac{9}{1024} = \frac{231}{1024} = 0.2256$$

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- 6 (a) Complete the table of values for $y = 1 + 3e^{-x}$ giving your answers to 2 decimal places where appropriate.

(2)

x	0	0.25	0.5	1	1.5	2	3
y	4	3.34	2.82	2.1	1.67	1.41	1.15

- (b) On the grid opposite, draw the graph of $y = 1 + 3e^{-x}$ for $0 \leq x \leq 3$

(2)

- (c) By drawing an appropriate straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

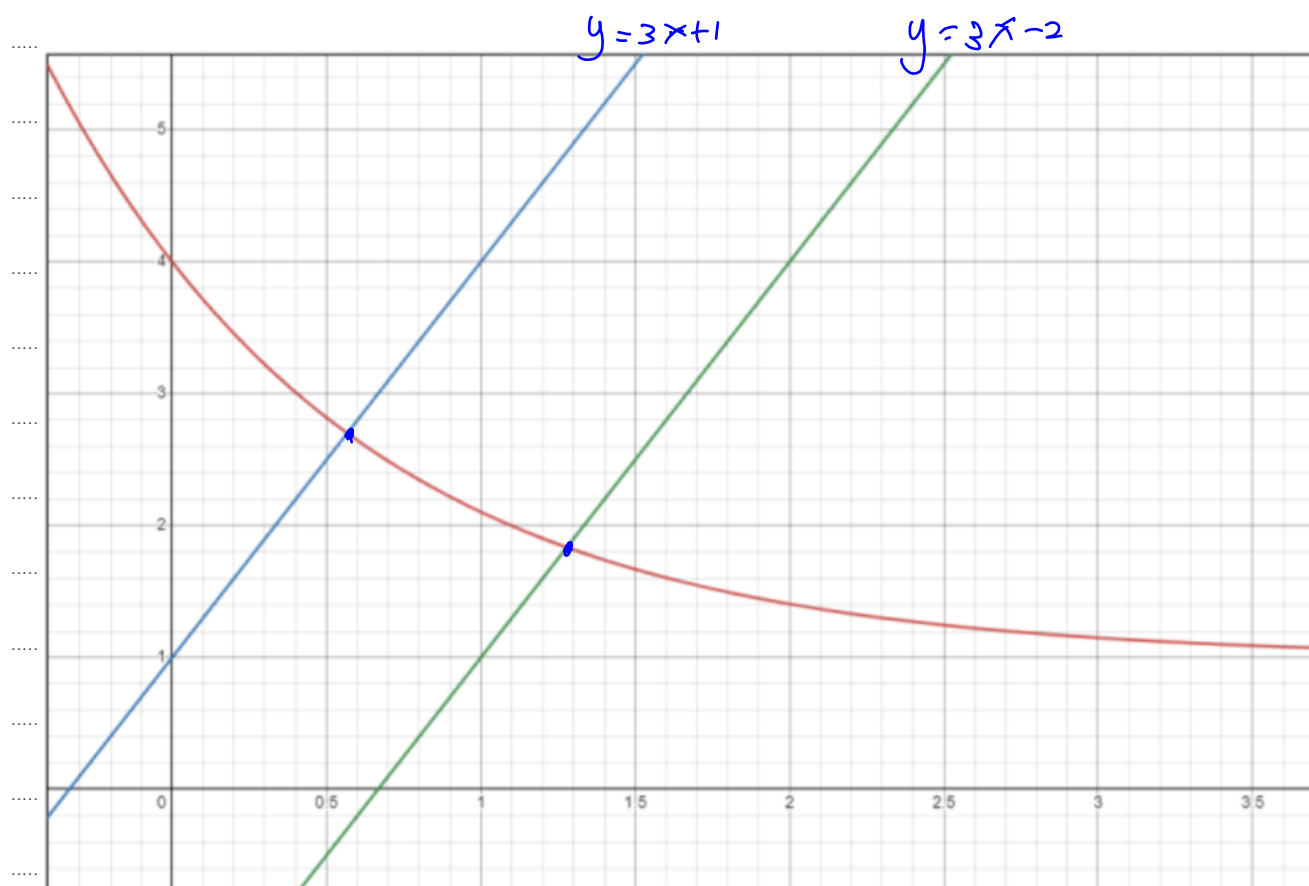
$$x = e^{-x} \quad \text{in the interval } 0 \leq x \leq 3$$

(3)

- (d) By drawing an appropriate straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$\ln(x-1)^3 = -3x \quad \text{in the interval } 0 \leq x \leq 3$$

(4)



Question 6 continued

$$(c) \quad x = e^{-x}$$

$$3x = 3e^{-x}$$

$$3x+1 = 1+3e^{-x}$$

$$y = 3x+1 \quad \Rightarrow \quad x = 0.5 \text{ (or } 0.6)$$

$$(d) \quad \ln(x-1)^3 = -3x$$

$$3\ln(x-1) = -3x$$

$$\ln(x-1) = -x$$

$$e^{-x} = x-1$$

$$3e^{-x} = 3x-3$$

$$1+3e^{-x} = 3x-2$$

$$y = 3x-2 \quad \leadsto \quad x = 1,2 \text{ (or } 1,3)$$

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7 The point with coordinates $(4, -104)$ lies on the curve C with equation $y = f(x)$

Given that $f'(x) = 4x^3 - 12x^2 - 19x + 12$

(a) (i) show that C passes through the origin,

(4)

(ii) show that C has a maximum at the point on the curve where $x = 0.5$

(3)

The curve C has another turning point at A and another turning point at B .

Given that the x coordinate of A is negative,

(b) (i) find the coordinates of A and the coordinates of B ,

(5)

(ii) determine the nature of these turning points.

(3)

$$\begin{aligned} \text{(a)(i)} \quad f(x) &= \int f'(x) \, dx \\ &= \int 4x^3 - 12x^2 - 19x + 12 \, dx \\ &= x^4 - 4x^3 - \frac{19}{2}x^2 + 12x + C \end{aligned}$$

$$f(4) = -104 \Rightarrow 4^4 - 4 \cdot 4^3 - \frac{19}{2} \cdot 4^2 + 12 \cdot 4 + C = -104$$

$$\Rightarrow C = 0$$

$$\Rightarrow f(0) = 0$$

$$\text{(ii)} \quad f'\left(\frac{1}{2}\right) = 4\left(\frac{1}{2}\right)^3 - 12\left(\frac{1}{2}\right)^2 - \frac{19}{2} + 12 = 0 \Rightarrow x = 0 \text{ is a stationary point}$$

$$\Rightarrow f''(x) = 12x^2 - 24x - 19 \quad 12\text{ANS}^2 - 24\text{ANS} - 19$$

$$f''\left(\frac{1}{2}\right) = 12\left(\frac{1}{2}\right)^2 - \frac{24}{2} - 19 = -28 < 0 \Rightarrow x = 0.5 \text{ is maximum}$$

(c)(i) $(x - \frac{1}{2})$ or $(2x - 1)$ is a factor of $f'(x) = 0$

$$\begin{array}{r|l} 4x^3 - 12x^2 - 19x + 12 & 2x - 1 \\ \hline 4x^3 - 2x^2 & \\ \hline -10x^2 - 19x & \\ -10x^2 + 5x & \\ \hline -24x + 12 & \\ -24x + 12 & \\ \hline 0 & \end{array} \quad \begin{aligned} &\Rightarrow f'(x) = (2x - 1)(2x^2 - 5x - 12) = 0 \\ &\Rightarrow (2x - 1)(x - 4)(2x + 3) = 0 \\ &\quad x = \frac{1}{2} \quad x = 4 \quad x = -\frac{3}{2} \\ &\Rightarrow A\left(-\frac{3}{2}, \frac{333}{16}\right), B(4, -104) \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad f''\left(-\frac{3}{2}\right) &= 44 > 0 \Rightarrow A \text{ is minimum} \\ f''(4) &= 77 > 0 \Rightarrow B \text{ is minimum} \end{aligned}$$



8 The volume of a sphere is 500 cm^3

(a) Calculate the radius, in cm to 3 significant figures, of the sphere.

(2)

The surface area of the sphere is increased by 20 cm^2

(b) Using calculus, find an estimate for the increase in the radius, in cm to 2 significant figures, of the sphere.

(5)

$$\begin{aligned} (a) \quad V &= \frac{4}{3}\pi r^3 = 500 \\ r^3 &= \frac{3 \cdot 500}{4\pi} = \frac{375}{\pi} \\ r &= \sqrt[3]{\frac{375}{\pi}} = 4.92 \end{aligned}$$

$$(b) \quad dA = 20$$

$$A = 4\pi r^2$$

$$\frac{dA}{dr} = 8\pi r$$

$$\Rightarrow dr = \frac{dA}{8\pi r} = \frac{20}{8\pi r} = \frac{5}{2\pi r}$$

dA : change in A

You can use:

$$\frac{dr}{dt} = \frac{dr}{dA} \cdot \frac{dA}{dt}$$

where $\frac{dA}{dt} = 20$

$$\text{When } r = 4.92 \Rightarrow dr = 0.16$$



9

$$f(x) = 3x^4 + 4x^3 - 36x^2 + 64$$

Given that $f(x)$ can be written in the form $(x-2)^2(ax^2 + bx + c)$

(a) find the value of a , the value of b and the value of c

(4)

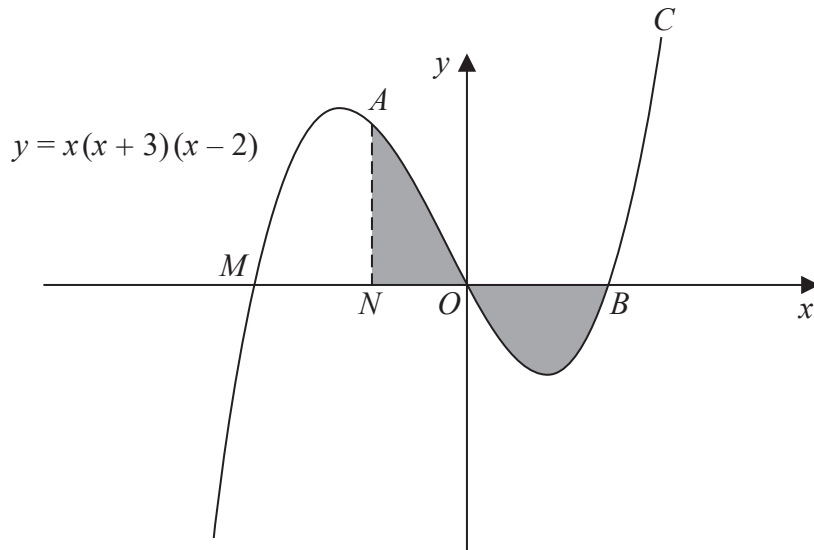


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Figure 3

Figure 3 shows a sketch of part of the curve C with equation $y = x(x+3)(x-2)$

The curve C crosses the x -axis at the point M , the origin and the point B .

The point N lies on the x -axis between M and O .

The point A lies on C such that AN is parallel to the y -axis.

The area of the shaded region bounded by the curve and OB is numerically equal to the area of the shaded region bounded by the curve, ON and NA .

Given that the coordinates of N are $(n, 0)$,

(b) use algebraic integration to show that n satisfies the equation

$$(x-2)^2(3x^2 + 16x + 16) = 0 \quad (7)$$

(c) Hence find the exact coordinates of A .

(5)

$$\begin{array}{r} (a) \quad 3x^4 + 4x^3 - 36x^2 + 64 \quad | \quad x^2 - 4x + 4 \\ \underline{3x^4 - 12x^3 + 12x^2} \\ 16x^3 - 48x^2 \\ \underline{16x^3 - 64x^2 + 64x} \\ 16x^2 - 64x + 64 \\ \underline{16x^2 - 64x + 64} \\ 0 \end{array}$$

$$\Rightarrow f(x) = (x-2)^2(3x^2 + 16x + 16)$$



Question 9 continued

(b) $f(x) = x^3 + x^2 - 6x$

$$A_1 = \int_0^2 x^3 + x^2 - 6x \, dx = \left[\frac{x^4}{4} + \frac{x^3}{3} - 3x^2 \right]_0^2$$

$$= \frac{2^4}{4} + \frac{2^3}{3} - 3 \cdot 2^2 = 4 + \frac{8}{3} - 12 = -\frac{16}{3}$$

$$A_2 = \int_n^0 x^3 + x^2 - 6x \, dx = \left[\frac{x^4}{4} + \frac{x^3}{3} - 3x^2 \right]_n^0$$

$$= -\frac{n^4}{4} - \frac{n^3}{3} + 3n^2$$

$$|A_1| = |A_2|$$

$$\Rightarrow -\frac{n^4}{4} - \frac{n^3}{3} + 3n^2 = \frac{16}{3}$$

$$\Rightarrow -3n^4 - 4n^3 + 36n^2 = 64$$

$$\Rightarrow 3n^4 + 4n^3 - 36n^2 + 64 = 0$$

$$\stackrel{(a)}{=} (n-2)^2(3n^2 + 16n + 16) = 0$$

So $x=n$ satisfies the equation

(c) $(n-2)^2(3n^2 + 16n + 16) = 0$

$$(n-2)^2(3n+4)(n+4) = 0$$

$$n=2 \quad \underline{n=-\frac{4}{3}} \quad n=-4$$

rej $n=-\frac{4}{3}$ rej $(-3 < n < 0)$

for $x = -\frac{4}{3} : y = \frac{200}{27}, A(-\frac{4}{3}, \frac{200}{27})$



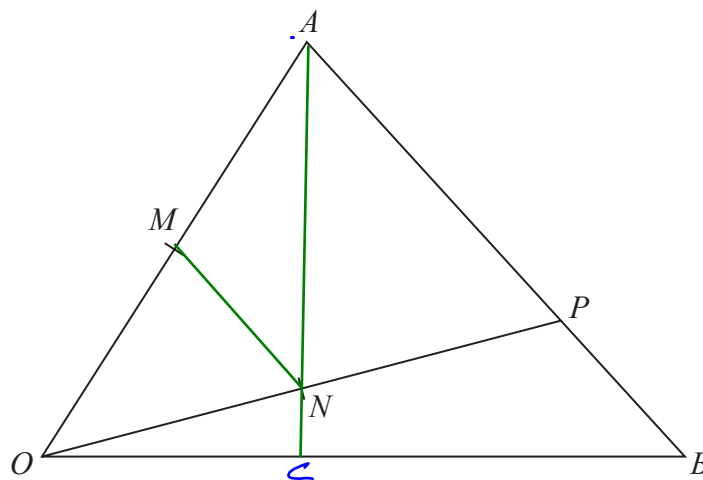


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Figure 4

Figure 4 shows triangle OAB in which

$$\vec{OA} = \mathbf{a} \text{ and } \vec{OB} = \mathbf{b}$$

The point P lies on AB such that $AP:PB = 3:1$

The point M is the midpoint of OA and the point N is the midpoint of OP .

(a) Find, as simplified expressions in terms of $\mathbf{a}$ and $\mathbf{b}$, the vector

(i) $\vec{OP}$ (ii) $\vec{MN}$

(4)

The point C lies on OB such that ANC is a straight line.

(b) Using a vector method, find the vector $\vec{OC}$ as a simplified expression in terms of $\mathbf{b}$

(6)

Given that $\frac{\text{area of quadrilateral } AMNP}{\text{area of triangle } OAB} = K$

(c) find the exact value of K

(4)

$$\begin{aligned} \text{(a)(i)} \quad \vec{OP} &= \vec{OA} + \vec{AP} = \mathbf{a} + \frac{3}{4} \vec{AB} \\ &= \mathbf{a} + \frac{3}{4} (\mathbf{b} - \mathbf{a}) = \frac{1}{4} \mathbf{a} + \frac{3}{4} \mathbf{b} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \vec{MN} &= \vec{MO} + \vec{ON} = -\vec{OM} + \vec{ON} = -\frac{1}{2} \vec{OA} + \frac{1}{2} \vec{OP} \\ &= -\frac{1}{2} \mathbf{a} + \frac{1}{8} \mathbf{a} + \frac{3}{8} \mathbf{b} = -\frac{3}{8} \mathbf{a} + \frac{3}{8} \mathbf{b} \end{aligned}$$



Question 10 continued

$$(b) \cdot \vec{OC} = \lambda \vec{OB} = \lambda b$$

$$\begin{aligned} \cdot \vec{OC} &= \vec{OA} + k\vec{AN} = \vec{OA} + k(\vec{AO} + \vec{ON}) \\ &= a - ka + \frac{k}{2} \cdot \left(\frac{a}{4} + \frac{3}{4}b \right) \\ &= a(1-k) + \frac{ak}{8} + \frac{3k}{8}b \\ &= a\left(1 - \frac{7k}{8}\right) + \frac{3k}{8}b \end{aligned}$$

$$\Rightarrow \lambda b = a\left(1 - \frac{7k}{8}\right) + \frac{3k}{8}b$$

$$\Rightarrow 1 - \frac{7k}{8} = 0 \Rightarrow \frac{7k}{8} = 1 \Rightarrow k = \frac{8}{7}$$

$$\Rightarrow \lambda = \frac{3k}{8} = \frac{3}{7} \Rightarrow \boxed{\vec{OC} = \frac{3}{7}b}$$

$$(c) \text{ Since } \vec{AP} = \frac{3}{4}\vec{AB}$$

$$\Rightarrow A_{OAP} = \frac{3}{4}A_{OAB}$$

$$\text{Since } \vec{OM} = \frac{1}{2}\vec{OA} \text{ and } \vec{ON} = \frac{1}{2}\vec{OP}$$

$$\Rightarrow A_{OMN} = \frac{1}{4}A_{OAP} = \frac{3}{16}A_{OAB}$$

$$\Rightarrow A_{AMNP} = A_{OAP} - A_{OMN} = \frac{9}{16}A_{OAP}$$

$$\Rightarrow \frac{A_{AMNP}}{A_{OAB}} = \frac{9}{16}$$